1. Zmodyfikowany kod z zajęć

```
function bspline_curve3D(precision, knot_vector, points)
if mod(precision, 2) == 0
   precision = precision + 1;
end
compute_nr_basis_functions = @(knot_vector, p) size(knot_vector, 2) - p - 1;
mesh = @(a, c) [a:(c-a)/precision:c];

p = compute_p(knot_vector);
t = check_sanity(knot_vector, p);

if (~t)
  disp("Poorly constructed knot_vector")
  return
end
nr = compute_nr_basis_functions(knot_vector, p);

[n, m] = size(points);

% Add 3rd dimension
if (n ~= nr || m ~= 3)
    disp("Poorly constructed points array")
    return
end

x_begin = knot_vector(1);
x_end = knot_vector(size(knot_vector, 2));
x = mesh(x_begin, x_end);

res_x = zeros(1, numel(x));
res_y = zeros(1, numel(x));
res_z = zeros(1, numel(x)); % Add 3rd dimension

for ind = 1:nr
    spline = compute_spline(knot_vector, p, ind, x);
    fprintf('\n');
    res_x = res_x + (spline .* points(ind, 1));
    res_y = res_y + (spline .* points(ind, 2));
    res_z = res_z + (spline .* points(ind, 3)); % Add 3rd dimension
end

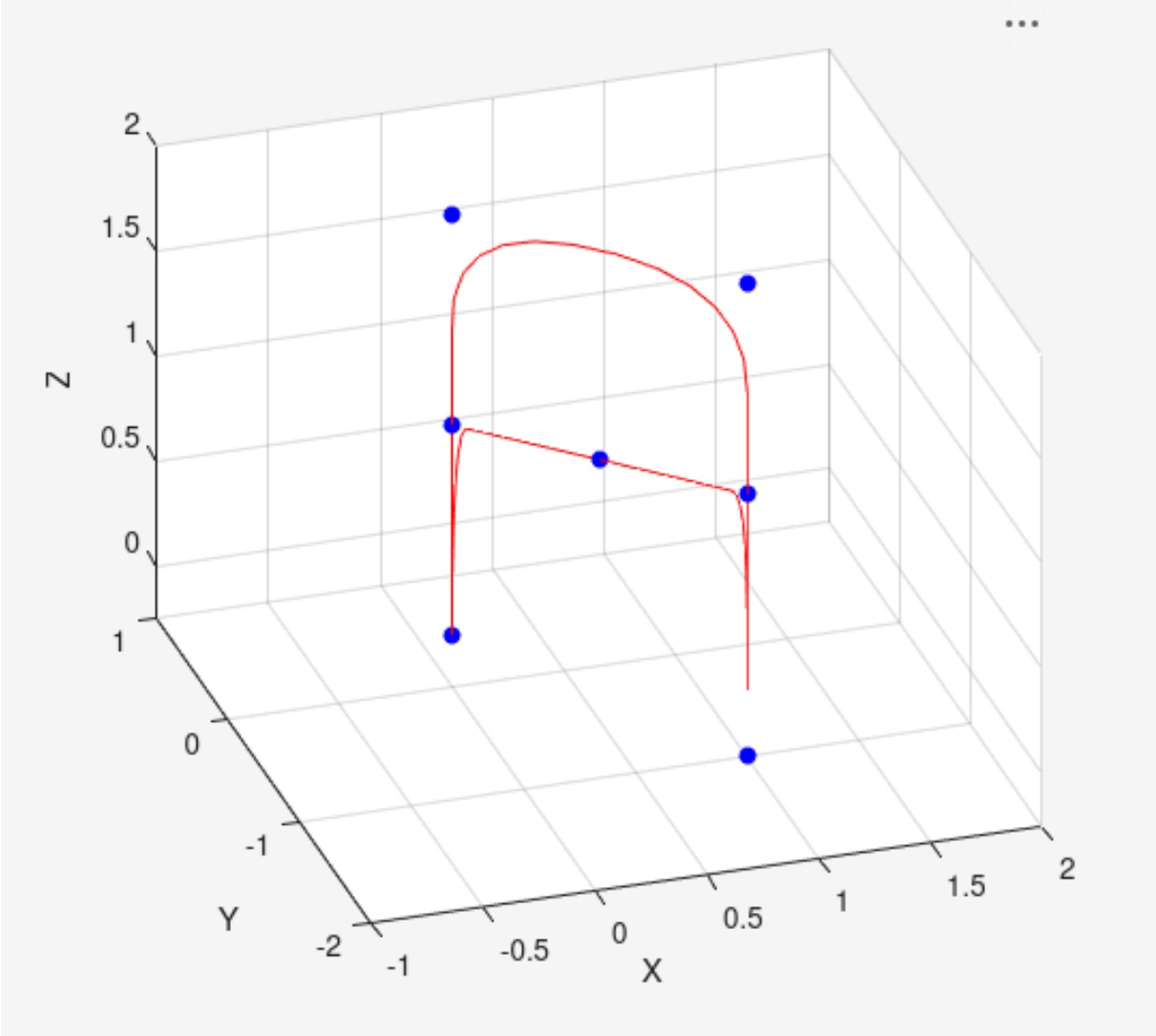
% Change plotting to 3D
plot3(res_x, res_y, res_z, 'r')
axis tight;
hold on
scatter3(points(:, 1), points(:, 2), points(:, 3), 'b', "filled")
grid on
axis equal
xlabel('X')
ylabel('Y')
zlabel('Z')
ylim([-2 1])
xlim([-1 2])
view(3)
hold off;
```

2. Rysowanie A

2.1. Wywołanie

```
bspline_curve3D(100, [0 0 0 1 2 3 4 5 6 15 16 16 16], [0 0 0; 0 0 1; 0 0 2; 1 -1 2; 1 -1 1; 1 -1 -0.25; 1 -1 1; 0.5 -0.5 1; 0 0 1; 0 0 0])
```

2.2. Wygenerowany wykres

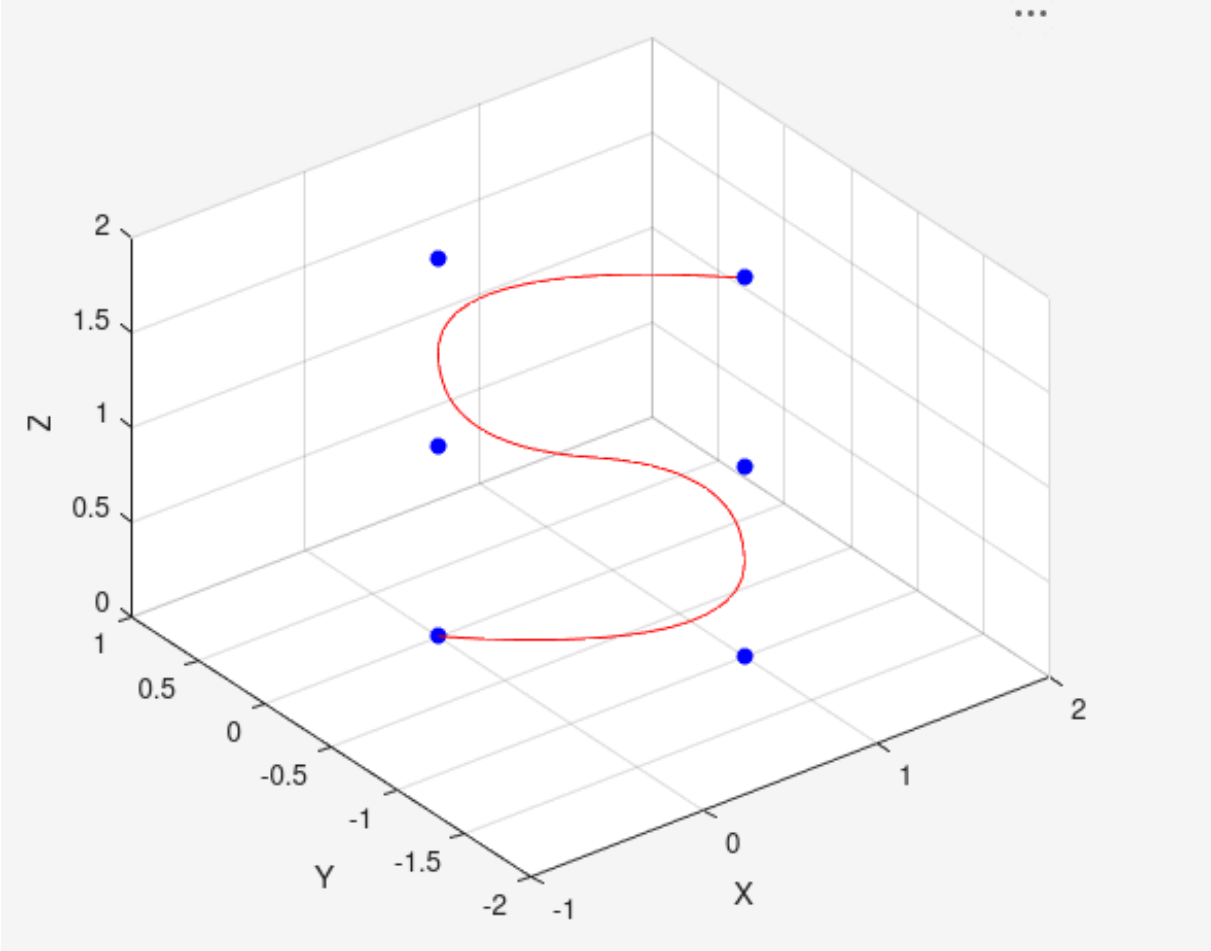


3. Rysowanie S

3.1. Wywołanie

```
bspline_curve3D(100, [0 0 0 1 2 3 4 4 4], [0 0 0; 1 -1 0; 1 -1 1; 0 0 1; 0 0 2; 1 -1 2])
```

3.2. Wygenerowany wykres



4. Rysowanie AS

4.1. Wywołanie

```
bspline_curve3D(100, [0 0 0 1 2 3 4 5 6 15 16 17 18 19 20 21 21 21], [1 -1 0; 1 -1 1; 1 -1 2; 0 0 2; 0 0 1; 0 0 -0.25; 0 0 1; 0.5 -0.5 1; 1 -1 1; 1 -1 0; 2 -2 0; 2 -2 1; 1 -1 1; 1 -1 2; 2 -2 2])
```

4.2. Wygenerowany wykres

